G-INVERSE AND G-SHRIEK IDENTIFY QCOH WITH QC MODULES ON THE LISSE-ETALE AND FLAT-FPPF SITES

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Editorial note. Spot-check note: The isomorphism g_!g^{-1}F -> F in (a) is concluded by 'the discussion in Sheaves on Stacks, Sections ... shows it is an isomorphism' - a section-discussion reference rather than a numbered lemma.

Allowed standard prerequisites (editorial).

- The lisse-etale and flat-fppf morphisms of topoi. Cohomology of Algebraic Stacks, Lemma stacks-cohomology-lemma-lisse-etale. g^* = g^{-1} and g^{-1}g_! = id on sheaves.
- Q-adjunction defect is parasitic. Cohomology of Algebraic Stacks, Lemma stacks-cohomology-lemma-adjoint-kernel-parasitic (2). Kernel/cokernel parasitic.
- Parasitic detection on the sites. Cohomology of Algebraic Stacks, Lemma stacks-cohomology-lemma-parasitic-in-terms-flat-fppf. g^* of a parasitic module is zero.

Source reference. Original lemma 0788: sources/stacks-chow/tags.txt, source lines 9228–9228. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0778: sources/stacks-chow/tags.txt, source lines 9193–9193. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03DO: sources/stacks-chow/tags.txt, source lines 4427–4427. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0EM8: sources/stacks-chow/tags.txt, source lines 18298–18298. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 07B0: sources/stacks-chow/tags.txt, source lines 9325–9325. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 06WS: sources/stacks-chow/tags.txt, source lines 8826–8826. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:stacks-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 06WU: sources/stacks-chow/tags.txt, source lines 8828–8828. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0779: sources/stacks-chow/tags.txt, source lines 9194–9194. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 07AR: sources/stacks-chow/tags.txt, source lines 9316–9316. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack.*

(1) *With g as in Lemma 0788 for the lisse-étale site we have*
   (a) *the functors g^{-1} and $g_!$ define mutually inverse functors*

$$QCoh(\mathcal{O}_\mathcal{X}) \underset{g_!}{\overset{g^{-1}}{\rightleftarrows}} QCoh(\mathcal{X}_{lisse,étale}, \mathcal{O}_{\mathcal{X}_{lisse,étale}})$$

   (b) *if $\mathcal{F}$ is in $LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$ then $g^{-1}\mathcal{F}$ is in $QCoh(\mathcal{O}_{\mathcal{X}_{lisse,étale}})$ and*
   (c) *$Q(\mathcal{F}) = g_!g^{-1}\mathcal{F}$ where Q is as in Lemma 0778.*

(2) *With g as in Lemma 0788 for the flat-fppf site we have*
   (a) *the functors g^{-1} and $g_!$ define mutually inverse functors*

$$QCoh(\mathcal{O}_\mathcal{X}) \underset{g_!}{\overset{g^{-1}}{\rightleftarrows}} QCoh(\mathcal{X}_{flat,fppf}, \mathcal{O}_{\mathcal{X}_{flat,fppf}})$$

   (b) *if $\mathcal{F}$ is in $LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$ then $g^{-1}\mathcal{F}$ is in $QCoh(\mathcal{O}_{\mathcal{X}_{flat,fppf}})$ and*
   (c) *$Q(\mathcal{F}) = g_!g^{-1}\mathcal{F}$ where Q is as in Lemma 0778.*

Proof. Pullback by any morphism of ringed topoi preserves categories of quasi-coherent modules, see Modules on Sites, Lemma 03DO. Hence g^{-1} preserves the categories of quasi-coherent modules; here we use that $QCoh(\mathcal{O}_\mathcal{X}) = QCoh(\mathcal{X}_{étale}, \mathcal{O}_\mathcal{X})$ by Sheaves on Stacks, Lemma 0EM8. The same is true for $g_!$ by Lemma 07B0. We know that $\mathcal{H} \to g^{-1}g_!\mathcal{H}$ is an isomorphism by Lemma 0788. Conversely, if $\mathcal{F}$ is in $QCoh(\mathcal{O}_\mathcal{X})$ then the map $g_!g^{-1}\mathcal{F} \to \mathcal{F}$ is a map of quasi-coherent modules on $\mathcal{X}$ whose restriction to any scheme smooth over $\mathcal{X}$ is an isomorphism. Then the discussion in Sheaves on Stacks, Sections 06WS and 06WU (comparing with quasi-coherent modules on presentations) shows it is an isomorphism. This proves (1)(a) and (2)(a).

Let $\mathcal{F}$ be an object of $LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$. By Lemma 0779 the kernel and cokernel of the map $Q(\mathcal{F}) \to \mathcal{F}$ are parasitic. Hence by Lemma 07AR and since $g^* = g^{-1}$ is exact, we conclude $g^*Q(\mathcal{F}) \to g^*\mathcal{F}$ is an isomorphism. Thus $g^*\mathcal{F}$ is quasi-coherent. This proves (1)(b) and (2)(b). Finally, (1)(c) and (2)(c) follow because $g_!g^*Q(\mathcal{F}) \to Q(\mathcal{F})$ is an isomorphism by our arguments above. □